

Nicholas Perello

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EDUCATION

University of Massachusetts Amherst

Ph.D. in Computer Science

Amherst, MA

September 2020 – Expected January 2027

Dissertation: *Towards Fair and Explainable Artificial Intelligence*

Advisors: Przemyslaw Grabowicz & Yair Zick

Admitted to the Ph.D. program starting Fall 2020.

University of Massachusetts Amherst

M.S. in Computer Science

Amherst, MA

September 2018 – May 2020

University of Massachusetts Amherst

B.S. in Computer Science; B.S. in Mathematics

Amherst, MA

September 2014 – May 2018

RESEARCH INTERESTS

Responsible AI • AI Safety • Explainable AI • LLM Evaluation • ML Observability

PUBLICATIONS

*Equal Contribution, †Equal Advising

In Submission

- **Nicholas Perello**, Przemyslaw A. Grabowicz, Yair Zick. Grilling BBQ: The Limits of Bias Benchmarking in LLM Safety Reporting. 2026.

Archival Publications

- **Nicholas Perello**, Cyrus Cousins, Yair Zick†, Przemyslaw A. Grabowicz†. Discrimination Induced by Algorithmic Recourse Objectives. In ACM Conference on Fairness, Accountability, and Transparency (ACM FAccT), 2025.
- JJ (Jenny) Hamer*, **Nicholas Perello***, Jake Valladares, Vignesh Viswanathan*, Yair Zick. Simple Steps to Success: A Method for Step-Based Counterfactual Explanations. In Transactions on Machine Learning Research (TMLR), 2024.
- Przemyslaw A. Grabowicz*, **Nicholas Perello***, Kenta Takatsu. Learning from Discriminatory Training Data. In AAAI/ACM Conference on Artificial Intelligence, Ethics, and Society (AIES), 2023.
- Przemyslaw A. Grabowicz, **Nicholas Perello**, and Aarshee Mishra. Marrying Fairness and Explainability in Supervised Learning. In ACM Conference on Fairness, Accountability, and Transparency (ACM FAccT), 2022.
- Weihao Tan*, David Koleczek*, Siddhant Pradhan*, **Nicholas Perello**, Vivek Chettiar, Nan Ma, Aaslesha Rajaram, Vishal Rohra, Soundar Srinivasan, H M Sajjad Hossain†, and Yash Chandak†. On Optimizing Interventions in Shared Autonomy. In Thirty-Sixth AAAI Conference on Artificial Intelligence (AAAI), 2022. *Also appeared at ICML Workshop on Human-AI Collaboration in Sequential Decision-Making, 2021.*

Non-Archival Publications

- Przemyslaw A. Grabowicz, **Nicholas Perello**, Yair Zick. Towards an AI Accountability Policy. In Comment on FR Doc # 2023-07776 Posted by the National Telecommunications and Information Administration, 2023.
- Aarshee Mishra*, **Nicholas Perello***, and Przemyslaw A. Grabowicz. Towards Fair and Explainable Supervised Learning. In ICML Workshop on Socially Responsible Machine Learning (SRML), 2021.
- Przemyslaw A. Grabowicz and **Nicholas Perello**. Resilience of Supervised Learning Algorithms to Discriminatory Poisoning of Training Data. In 7th International Conference on Computational Social Science (IC2S2), 2021.

AWARDS

- GEM Ph.D. Engineering & Science Fellowship (awarded Fall 2020).
- UMass Amherst 2022-23 Distinguished Teaching Award Finalist – Highest recognition of teaching awarded at UMass Amherst, finalist from the Manning College of Information and Computer Science.
- UMass Amherst 2022-23 Commitment to Diversity Award Winner – Awarded to one student for their outstanding contributions to diversity and inclusion within UMass Amherst.

RESEARCH EXPERIENCE

Socially Intelligent Media & Systems Lab and Fair & Explainable Decision Making Lab

UMass Amherst

Advisors: Przemysław Grabowicz & Yair Zick

September 2020 – Present

- Dissertation research on fairness, explainability, and their intersection in machine learning.
- Current projects use red teaming, bias benchmarking, and mechanistic interpretability to evaluate and explain the behavior of large language models (LLMs).
- With MassMutual: designing a benchmark for auditing LLM self-explanations, evaluating self-explanation faithfulness and concealment of spurious-feature influence in decision-making.
- Additional explainable AI research collaborations with Adobe and US Army DEVCOM.
- Authored posts on fairness and explainability for the UMass [CICS EQUATE and Machine Learning Blog](#).

UMass – Google DeepMind Collaboration

UMass Amherst

Collaborator: JJ Hamer

September 2023 – Present

- Developed a data-driven algorithmic recourse method (TMLR, 2024) and an accompanying Python framework implementing it alongside state-of-the-art recourse methods.
- Currently studying self-recognition bias in LLM evaluators under persona-conditioned generation for application in multi-agent settings.

UMass – Microsoft (MAIDAP) Collaboration

UMass Amherst

Collaboration Lead: H M Sajjad Hossain

January 2020 – April 2022

- Collaborated with the Microsoft AI Development Acceleration Program on the development and study of intervention-aware multi-agent reinforcement learning algorithms for assisting individuals with disabilities in games without degrading their user experience (AAAI, 2022).

UMass Center for Data Science

UMass Amherst

Program Heads: Brant Cheikes & Matthew Rattigan

June 2019 – August 2019

- Partnered with Springfield Public Schools to aid in their investigation of the diminishing college success rates of their students using exploratory data analysis and machine learning with students' academic histories.

Laboratory for Advanced Software Engineering Research

UMass Amherst

Advisor: Leon Osterweil

December 2016 – May 2018

- Investigated methods to reduce the risk of errors in complex medical procedures by assisting medical staff with in-house process models and simulations developed collaboratively with domain experts.

INDUSTRY EXPERIENCE

Salesforce

San Francisco, CA

Software Engineering Intern

June 2018 – August 2018

- Full stack developer of a Service Cloud Chatbot feature that streamlined the retrieval of knowledge articles by removing the need for clients to develop code for this task for each of their deployed chatbots.

Intel Corporation

Hillsboro, OR

Software Engineering Intern

June 2017 – August 2017

- Given ownership of internal benchmark data analytics tool and extended it to enhance user experience, support more types of data, and present additional visualizations of data.

TEACHING EXPERIENCE

University of Massachusetts Amherst

Instructor of Record & Teaching Assistant

Amherst, MA

September 2020 – Present

Instructor of Record:

- **FYS191: First-Year Seminar** (Fall 2022, Fall 2023, Fall 2025)
- **CICS290C: Computing Success Strategies** (Fall 2023)

Teaching Assistant:

- **CS515: Algorithms, Game Theory, & Fairness** (Fall 2024)
- **CS383: Artificial Intelligence** (Spring 2021, Fall 2021, Spring 2022, Fall 2022)
- **CS345: Practice and Applications of Data Management** (Fall 2020)
- **CICS110: Foundations of Programming** (Spring 2024)
- **CICS160: Object Oriented Programming** (Spring 2023)
- **INFO101: Introduction to Informatics** (Fall 2025)

SERVICE, MENTORSHIP, & OUTREACH

- **Code2040 Fellow and Senior Fellow** during the summers of 2017 and 2018.
- **Reviewing:** FORC 2024, WWW 2026, JAIR 2026.

University of Massachusetts Amherst – Manning College of Information & Computer Science

- **Co-founder of B[U]ILT (Black, Indigenous, and Latinx in Tech)** (Co-chair Fall 2018 – Fall 2021).
- **Co-chair of CARE (Committee Against Racism and for Equity) Inclusive Teaching Subcommittee** (Summer 2020 – Fall 2021).
- **Member of Teaching Development Committee** (Fall 2021 – Spring 2022).
- **UGRAD Research Volunteer (URV) Program Ph.D. Mentor** (Winter 2020-21).
- **Energizing Mentoring and Broadening Exposure to Research (EMBER) Program Ph.D. Mentor** (Spring 2021).

PROGRAMMING SKILLS

Python • PyTorch • HuggingFace/Transformers • Pandas • NumPy • scikit-learn • SQL • Java • Git